Stochastic Modelling Project 1

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Abstract

This report applies stochastic modelling techniques in two contexts: the spread of infectious diseases and the occurrence of insurance claims. In the first part, we construct and analyse a Markov chain model for the spread of measles, incorporating state transitions between susceptible, infected, and recovered individuals. We examine the system's long-run behaviour, investigate time-varying infection probabilities, and explore the impact of vaccination through simulation. In the second part, we model insurance claim arrivals using a Poisson process and claim sizes using an exponential distribution. Analytical results are then compared with results from Monte Carlo simulations.

1 Problem 1: Modelling the Outbreak of Measles

1.1 Context and Setup

The first problem investigates a simplified stochastic model for the spread of measles. Each individual in a fixed population of size N can, according to given probabilities each day, transition between the three states: susceptible (S), infected (I), and recovered (R).

We model on a daily scale and let $n = 0, 1, 2, \ldots$ denote time measured in days. Each day:

1. a susceptible individual can become infected or remain susceptible,
2. an infected individual can become recovered or remain infected,
3. a recovered individual can lose immunity and become susceptible, or remain recovered.

In the first stage of modelling, we assume that individuals in the population are independent, and that each day:

- a susceptible individual has probability $0 < \beta < 1$ of becoming infected,
- an infected individual has probability $0 < \gamma < 1$ of becoming recovered,
- a recovered individual has probability $0 < \alpha < 1$ of losing immunity.

1.2 Modelling a Single Individual

Consider a specific individual, and let X_n denote their state at time n. Let states 0, 1, and 2 correspond to S, I, and R, respectively, and let $\{X_n : n = 0, 1, 2, \ldots\}$ be a stochastic process with realizations $\{x_0, x_1, \ldots\}$.

Given $X_n = i$, we have:

1. if $i = 0$, then $P(X_{n+1} = 1) = \beta$, $P(X_{n+1} = 0) = 1 - \beta$;
2. if $i = 1$, then $P(X_{n+1} = 2) = \gamma$, $P(X_{n+1} = 1) = 1 - \gamma$;
3. if $i = 2$, then $P(X_{n+1} = 0) = \alpha$, $P(X_{n+1} = 2) = 1 - \alpha$.

These probabilities are independent of previous states $X_{n-1}, X_{n-2}, \ldots, X_0$. Hence, $\{X_n\}$ satisfies the Markov property and is a Markov chain with transition matrix:

$$\mathbf{P} = \begin{bmatrix} 1-\beta & \beta & 0 \\ 0 & 1-\gamma & \gamma \\ \alpha & 0 & 1-\alpha \end{bmatrix}.$$

1.3 Mean Time Spent in Each State

Since the state space is finite and $\mathbf{P}$ is regular (i.e. some power $\mathbf{P}^k$, such as $\mathbf{P}^2$, has all positive entries), the chain has a limiting distribution $\boldsymbol{\pi} = (\pi_0, \pi_1, \pi_2)^\intercal$ satisfying:

$$\boldsymbol{\pi} = \boldsymbol{\pi}\mathbf{P}, \qquad \sum_{i=0}^{2} \pi_i = 1.$$

For $\beta = 0.01$, $\gamma = 0.10$, and $\alpha = 0.005$, solving gives:

$$\boldsymbol{\pi} = \left(\frac{10}{31}, \frac{1}{31}, \frac{20}{31}\right)^\intercal.$$

Hence, the long-run mean number of days per year spent in each state is:

$$365 \times \boldsymbol{\pi} = (117.74, 11.77, 235.48)^T.$$

We verify with 30 simulations of the stochastic process. The approximate 95% confidence intervals for these quantities are:

$$\pi_0 \in [100.407, 118.686], \quad \pi_1 \in [10.261, 12.486], \quad \pi_2 \in [234.575, 253.585].$$

The CIs are calculated by normal assumption according to the Central Limit Theorem, which is valid for large samples ($n \geq 30$). We can see that the confidence intervals are compatible with the exact calculation, although the they are not centred around the exact values for π_0 and π_2.

1.4 Varying Infection Rate

Due to the infectious nature of measles, β should realistically vary with the number of infected individuals. Assume that the population size $N = 1000$ is fixed and that at each time step there are S_n, I_n, and R_n susceptible, infected, and recovered individuals respectively. Define:

$$\beta_n = \frac{0.5 I_n}{N}, \qquad \gamma = 0.10, \qquad \alpha = 0.005,$$

such that at every step each individual transitions independently according to $(\beta_n, \gamma, \alpha)$.

Finally, let:

$$Y_n = (S_n, I_n, R_n), \qquad Z_n = (S_n, I_n).$$

We should now ask whether $\{I_n : n = 0, 1, ...\}$, $\{Z_n : n = 0, 1, ...\}$, and $\{Y_n : n = 0, 1, ...\}$ are Markov chains. Let $i < j < N$. Because $S_n + I_n + R_n = N$, we observe that $P(I_{n+1} = j \mid I_n = i, I_{n-1} = N) = 0$, since at time n there are 0 susceptible individuals and $N - i$ recovered individuals, making it is impossible for any more individuals to become infected at time $n + 1$. Clearly, $P(I_{n+1} = j \mid I_n = i) > 0$ when we have no further information on S_n. Thus, $P(I_{n+1} = j \mid I_n = i_n, I_{n-1} = i_{n-1}) \neq P(I_{n+1} = j \mid I_n = i_n)$, and $\{I_n\}$ is *not* a Markov chain.

To explain why, we observe that the event of an individual entering state I is a Bernoulli trial with probability β_n, and that the event of an individual leaving state I is a Bernoulli trial with probability γ. If ΔI_{n+1} is the change in number of infected individuals between time $n + 1$ and n, then this implies that

$$\Delta I_{n+1} = B_S - B_I$$

where $B_s \sim \text{Binomial}(S_n, \beta_n)$ and $B_I \sim \text{Binomial}(I_n, \gamma)$. The transition probability from state i to j is then $P(B_S - B_R = j - i)$. We cannot simply calculate these transition probabilities as S_n is a random variable dependent on the previous states of $\{I_n\}$.

With $\{Z_n\}$ on the other hand, the transition probability from state (s_n, i_n) to (s_{n+1}, i_{n+1}) is

$$P(B_S - B_I = i_{n+1} - i_n, B_R - B_S = s_{n+1} - s_n)$$

where $B_R \sim \text{Binomial}(N - I_n - S_n, \alpha)$. Here, since both I_n and S_n are known, these transition probabilities are well defined and dependent only on the current state Z_n. Therefore, we can conclude that while $\{I_n\}$ is not a Markov chain, both $\{Z_n\}$ and $\{Y_n\}$. This arises from the fact that the latter contains all the information required to evolve the system in a single state, while the former does not.

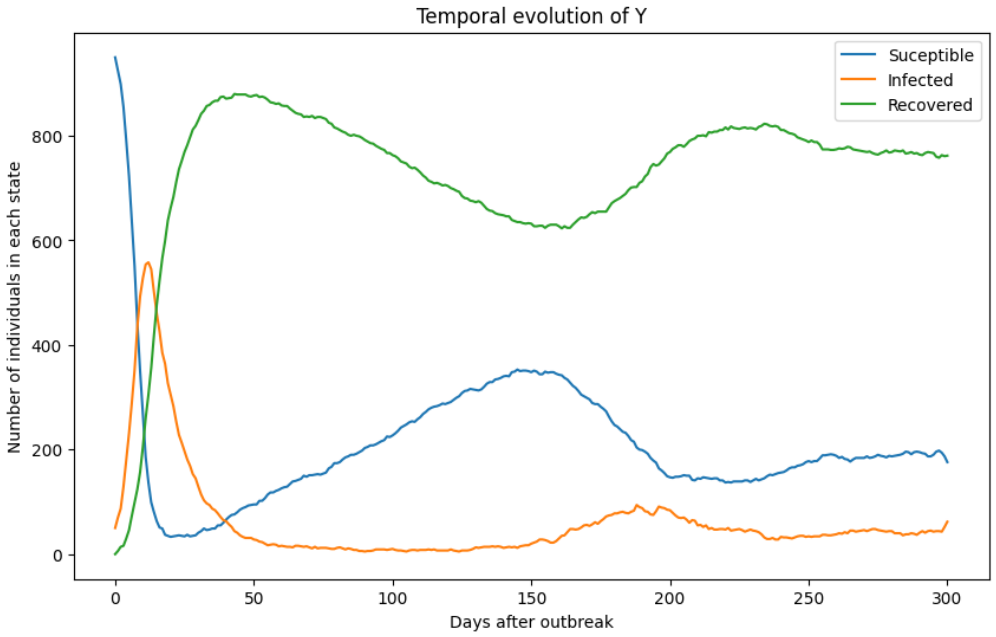


Figure 1: Temporal evolution of one realization of the measles outbreak for 1000 individuals.

Figure 1 shows one realization of $\{Y_n\}$ for $n \leq 300$. We can see that the behaviour is quite different in the time intervals 0-50 and 50-300. At the start of the outbreak, nearly the entire population is susceptible to the disease, and in the following days the infection spreads exponentially fast as a consequence, and virtually everybody gets infected. This continues until the susceptible population is mostly exhausted, at which point the infection rate drops to nearly zero and the infection reaches its peak. The number of infected drops in the subsequent days as infected individuals recover. Right after the first wave, the majority of the population is immune, and there are few susceptible individuals for the disease to spread to. The number of infected begins to drop until it is close to zero.

In the next stage, for days 50-300, the system experiences smaller recurring waves. The cycle goes as follows: after the previous wave has has subsided, nearly the entire population is immune, so the disease does not spread and there are few infected individuals. As time passes and immune individuals become susceptible again, the infection starts to spread, driving the infection rate up, and leading to a new wave. However, this wave is smaller than the first since there remains a significant number of immune individuals.

1.5 Explosive Behaviour

A key interest is the explosive behaviour during an outbreak, like the one we see in figure 1. The number of infected peaks around $n \approx 10$. We estimate:

$$E\left[\max_{n \leq 300}\{I_n\}\right] \quad \text{and} \quad E\left[\arg\max_{n \leq 300}\{I_n\}\right].$$

From 1000 simulations, the approximated 95% CIs are:

$$[522.846, 525.046] \text{ and } [11.850, 11.940]$$

respectively. Intuitively, these quantities can be used to determine the severity of a hypothetical outbreak. They represent the largest number of simultaneously infected individuals during the 300-day simulation window, and the time at which the outbreak is expected to reach its maximum. We see that for the parameters we have used, the infection takes only about 12 days to reach its peak, at which point over half the population will be infected.

1.6 Vaccination

One way of mitigating the spread of the disease is through vaccination. Let's examine how the dynamics change when we introduce vaccination and lifelong immunity. We consider $V = 100$, 600, and 800 vaccinated individuals, out of $N = 1000$, and introduce 50 infected individuals among the unvaccinated. Since vaccinated individuals are completely immune, they effectively reduce β_n by lowering the susceptible fraction. Thus, we keep N constant while setting $S_0 = N - V - 50$ (i.e. add a trivial state V with probability 0 of leaving and 0 of entering).

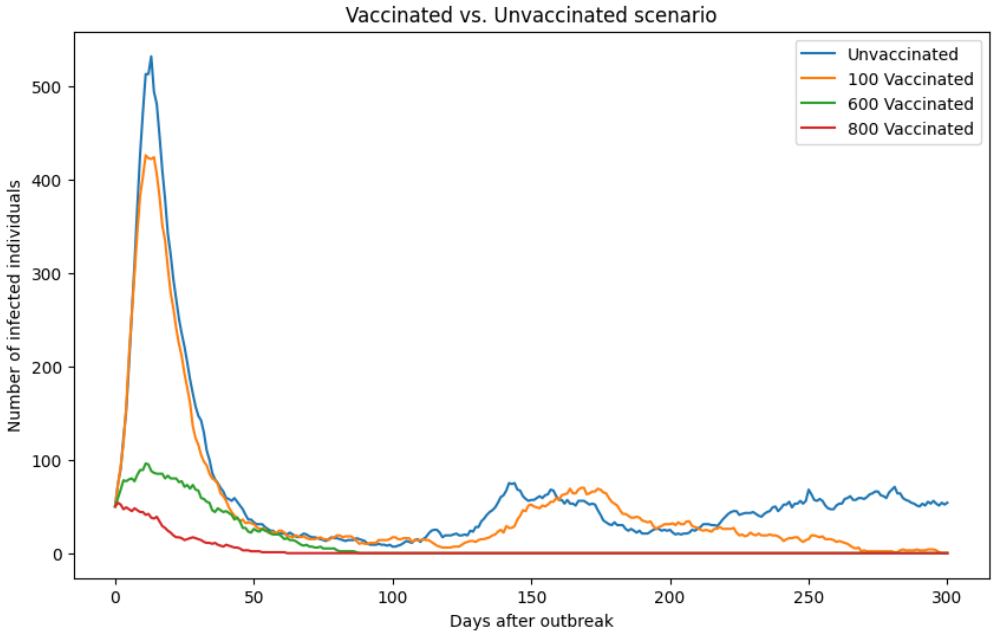


Figure 2: Temporal evolution of the measles outbreak with varying numbers of vaccinated individuals.

Table 1: Relevant quantities

Number of vaccinated individuals	95% CI for peak number of infected	95% CI for time of peak
0	$[522.846, 525.046]$	$[11.850, 11.940]$
100	$[438.879, 440.933]$	$[12.608, 12.714]$
600	$[95.718, 97.076]$	$[15.434, 15.904]$
800	$[51.250, 51.488]$	$[0.952, 1.144]$

From Figure 2 and Table 1 we observe that:

1. The peak number of infected decreases with more vaccinated individuals.

2. Except for a very high number of vaccinated, the time until peak increases with vaccination.

3. For high numbers of vaccinated individuals, the disease quickly dies out.

In fact, when 80% of the population is vaccinated, the disease effectively does not spread beyond the initial number of infected. The infection rate is so low in these scenarios that the number of infected eventually reaches 0, and since the infection rate is proportional to the number of infected, no new individuals can become infected and the disease is completely eradicated.

2 Problem 2: Insurance Claims

This problem models the arrival and size of insurance claims using Poisson and exponential distributions. The model is used to study both the number of claims and the total monetary claim amounts within a given period.

2.1 Number of Claims

Let $X(t)$ denote the number of claims received by an insurance company in time interval $[0, t]$. We will assume that $\{X(t) : t \geq 0\}$, where t is the number of days since January 1st, can be modelled as a Poisson process with rate $\lambda(t) = 1.5$ claims/day.

We compute the probability of more than 100 claims by March 1st (59 days):

$$P(X(59) > 100) = 1 - \sum_{x=0}^{100} e^{-1.5\times 59} \frac{(1.5 \times 59)^x}{x!} \approx 0.103.$$

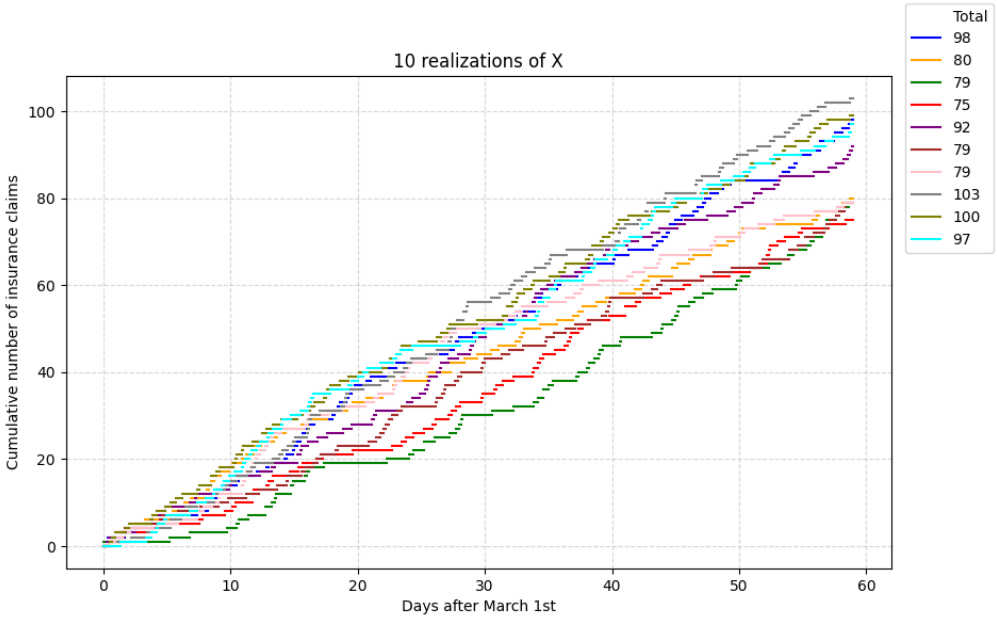


Figure 3: Ten simulations of the number of claims received between January 1st and March 1st.

From 1000 simulations, we get the approximation $P(X(59) > 100) = 0.098$ with a 95% CI of $[0.083, 0.113]$. The relatively wide interval despite high number of simulations indicates a high variance in the process.

2.2 Size of Monetary Claims

Let each claim amount C_i (in million kr.) follow an exponential distribution with rate $\gamma = 10$. Assume claim sizes are independent and independent of arrival times.

Then, the total claim amount by time t is:

$$Z(t) = \sum_{i=1}^{X(t)} C_i$$

And $Z(t) \mid (X(t) = x) \sim \Gamma(x, \gamma)$.

From 1000 simulations, the probability that $Z(59) > 8$ million kr. is 0.743 with a 95% CI of $[0.720, 0.766]$.

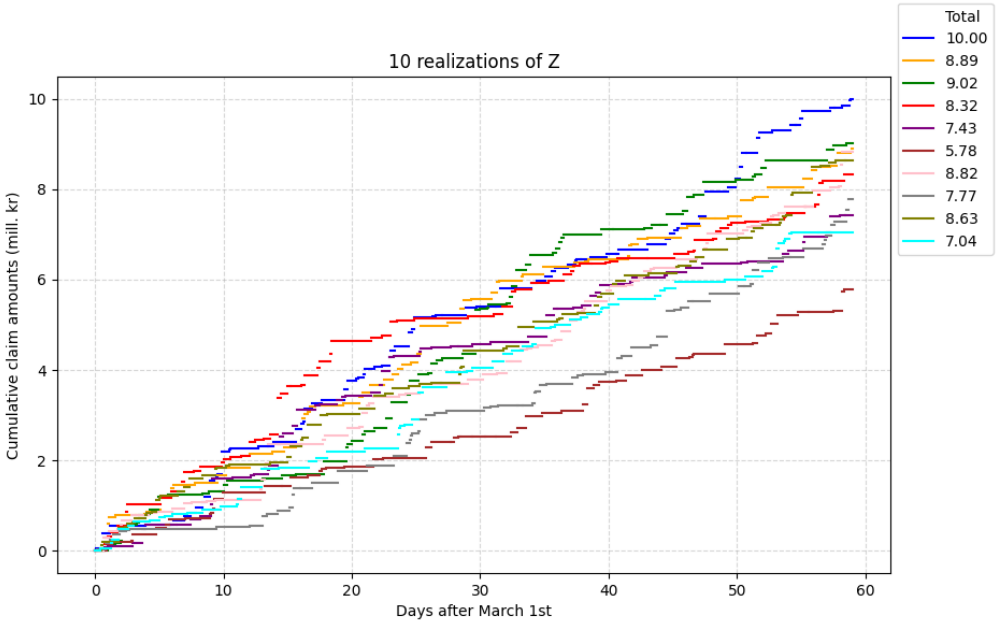


Figure 4: Ten simulations of total claim amounts between January 1 and March 1.

2.3 Investigating Claims

A claim is investigated if it exceeds 250 000 kr. For $C_i \sim \text{Exp}(\gamma)$, we find:

$$P(C_i > 0.25) = e^{-10 \times 0.25} \approx 0.082.$$

Thus, the number of claims exceeding this threshold in $[0, t]$, denoted Y_t, follows a binomial distribution:

$$Y_t \mid (X(t) = x) \sim \text{Binomial}(x, 0.082).$$

By the thinning property of Poisson processes:

$$Y_t \sim \text{Poisson}(0.082\lambda t) = \text{Poisson}(0.123t).$$